

With the growth in interest in quantum computing along with the rise of AI, simulating quantum computations has come to the forefront, because of the current unreliability of physical quantum computers.

The impetus for this project is the fact that qubit states grow at a rate of 2^n , rather than in classical bits, that grow at $2 \cdot n$. This is due to the required superposition necessary for each qubit. You need n pieces of information in classical computing, as opposed to 2^n pieces of information in quantum computing. This can cause quantum calculations to get out of hand extremely quickly.

There is a limited number of qubits in the world, and doing a large amount of quantum testing is difficult. Because of this, many people have turned to simulators online. This can be even more difficult and can take a long time because it requires quantum calculations and algorithms on a classical computer. Moving this to a hardware accelerator can allow for faster advancement and testing of quantum algorithms. The utility in this is that students and researchers will be able to quickly learn about the nature of quantum algorithms in practice without needing to rely on inaccessible quantum hardware.

We will be creating a hardware accelerator in SystemVerilog that will check and replace if there is a bit flip or a phase flip error in the quantum algorithm (Shor's error correcting algorithm). We are expecting to understand more about quantum simulators and how advancing the simulators can eventually lead to advancements in quantum algorithms. This project will design matrix multiplication, exponential, and eigenvalue modules in SystemVerilog and synthesize them into a quantum error detection and correcting algorithm. Finding and correcting errors in quantum computing is important and helpful, but only up to 3 qubits because of the probability that only one of the bits or phases will flip. This is still a useful algorithm because it can provide meaningful insight for the future of quantum simulating hardware accelerators.